

LLMs in the Shadow of Whirling Dervishes: The Overestimation of AI Creativity

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“He had effortlessly learned English, French, Portuguese, and Latin. I suspect, nevertheless, that he was not very good at thinking. To think is to ignore (or forget) differences, to generalize, to abstract. In the teeming world of Ireneo Funes there was nothing but particulars—and they were virtually immediate particulars.”

(Jorge Luis Borges, “Funes the Memorious”)

Borges’ protagonist in “Funes the Memorious” has an infallible memory, allowing him to make use of even the most trivial of memories when he thinks, yet Borges’ narrator comments, as an aside, that “thinking” is not something he was good at. The narrator’s characterization of Funes is striking with respect to large language models (LLMs). A real-life Funes is the subject of neuropsychologist Alexander Luria’s *The Mind of a Mnemonist*. At one point, Luria writes that S., the name he gives to the subject of his study, is called upon to memorize a sequence of numbers: 1234, 2345, 3456, etc. Though his recollection proved unblemished due to his extraordinary visual memory, he failed to notice the basic pattern. The ability to generalize is an intellectual imperative altogether separate from breadth. Unlike S. and other human savants, LLMs are precisely suited for such feats. This is no doubt a profound part of the human intellect that is now being outsourced. But human creativity cuts deeper, and as we will see, narrower.

Human brains have a severely limited operative storage capacity that is nevertheless well-indexed. The number of neurons and synapses humans possess peaks shortly after birth; the connections are abundant and diffuse, such that in time, many of them are pruned while the experientially propitious ones are strengthened and consolidated into adulthood. The Hebbian principle, that “neurons that fire together wire together,” is both a neurological explanation and a metaphor for human intelligence: more (diffuse

connections) is less if the less (compact connections) is more substantive. It demonstrates that human intelligence lies in specialization and deliberate narrowing of scope—in a capacity to generate finite, bounded, and coherent associations.

Psychologist Susan Engel shows that curiosity develops as such: young children enter with a porous and total curiosity towards the world, which then yields to an interest in narrower strands of experience. In her words, “curiosity becomes a narrower but perhaps brighter light, aimed with more intention;” and as the general quotidian order becomes more predictable, attention shifts to “details in a script, rather than the script itself.” In other words, we become more invested in the fine-grained and destabilizing anecdotes because most phenomena cease to surprise us at a certain point. The fact that curiosity might induce one person to learn about automobiles, another textiles, and yet another Victorian literature, is necessary in scaffolding a creative voice.

To use another Borges example, the LLMs act as prodigious bookkeepers of the Library of Babel, which, as described in the eponymous short story, is a seemingly endless storage facility where meaningful and meaningless books are indistinguishable from each other, given their identical façades. LLMs have a vast storage at their disposal, but unless one directs them to the vicinity of a certain text with precise or roundabout questions, they cannot harness the knowledge at their disposal. The LLM is a functionary to the human mastermind. It is the antithesis of the mature human brain, with its reduced complexity paired with specialized connections that selectively control for redundancy. But what does having such prodigious functionaries—research assistants, if you will—imply for the future of human creativity?

Well, for one, there will be fewer harbingers of the said virtue. If students approach schoolwork with the pragmatism of a business executive churning through

paperwork rather than the sustained probing of an inquirer, they will sever their birthright to creativity, as well as the experience of deep learning (alternately called expertise). This has been substantiated empirically by the recent oft-circulating MIT Media Lab study “Your Brain on ChatGPT,” where the brain activity of subjects prompted to write an essay showed a significantly negative correlation with ChatGPT use. These results no doubt herald a certain social regression, a deferral of the mores of literacy for a glitzy, inchoate technology, which constitutes an intellectual (in addition to economic) “AI Bubble.”

Take programming, where various menial coding jobs are presently being nullified; and yet, creativity has only been further invigorated by the encroachment of AI, because people can now innovate without knowing coding languages, and meanwhile, the higher rungs of expertise in the field have only gained in pertinence.

Just as non-coders are now “vibe coding”—tossing qualitative ideas at LLMs and having them do the technical labor—there will be “vibe readers” galore. The writer Joshua Rothman laments that his publications may come to be superseded by algorithmic distillations of original texts, hence a troubling separation between form and content: “As a writer, I may not want to see my text refracted in this way. But the power of refraction won’t be mine to control; it will lie with readers and their A.I.s. Together, they will collapse the space between reading and editing.”

Convoluting texts can be made more palatable because they effectively become mere means of arriving at one’s preferred medium/mode of learning, be it a condensed summary, a conversation, a podcast, etc. Increasingly, writers will also have LLMs in mind as important “readers” – if one can call them that – especially those publishing in relative obscurity and esoteria. Rothman eloquently concludes as such: “Text may get treated like a transitional medium, a temporary resting place for ideas. A piece of writing, which today is often seen as an end point, a culmination, a finished unit of effort, may, for better and worse, be experienced as a stepping stone to something else.” The transformation Rothman describes is one of formlessness, where content can take any number of configurations, none of which settle into permanence. But this is a continuity in intellectual history, not a paradigm-shifting rupture

Mechanisms of Human Cognition

In *Gesture and Speech*, archaeologist André Leroi-Gourhan theorizes the origins of human language. He

posits a gradual, modular process of externalizing physical and cognitive ability, which eventually culminates, if ever, in the exteriorization of sentience. Leroi-Gourhan implicates the advent of bipedalism (erect posture) in the development of language. Bipedalism freed our hands, resulting, first, in the functional shift of a newly liberated anatomical tool to be employed in the pursuit of fine motor demands, and second, the commencement of an evolutionary process that laid the foundation for human language; the freeing of hands led to the development of tools, then gesture, and finally, language, which led to and was reinforced by structural changes in the brain. Language allows us to bring our thoughts into the surrounding environment, and in turn, with every augmentation of informational storage capacity, ideas and experiences become part of a vast social apparatus constituting an expansive cultural memory.

Andy Clark and David Chalmers present a radical elaboration of human development in their seminal paper “The Extended Mind,” where they implicate objects extraneous to ourselves in the mechanisms of cognition. In other words, we incorporate objects into our computations – the process of, say, long multiplication entails assimilating the pen and paper into the substrate of our thoughts, manipulating symbols and concepts in a manner that is inseparable from these tools.

Crucially, the human computational model tends outwardly and inwardly in turn. Hermann von Helmholtz wrote, in a seminal late nineteenth-century work, that due to the sparsity of sensorial information, there must be (unconscious) inferences that aid in our interpretation of external reality. We don’t, then, absorb information in totalities. His medieval Arab predecessor, Ibn al-Haytham (also known as Alhazen), similarly challenged the prevailing theory of “extramission” in optics, proposing instead one of “intromission.” He suggested not only that physical rays reach the eye and make an imprint on it (rather than beams of light sent from the eye into the world returning with information), but also that the process of seeing involves an interpretive process of discernment that happens in our cognition. The abundance of neurons in the primary visual cortex illustrates the mental effort required to make sense of information from the retina. Rodrigo Quian Quiroga writes that “the brain does not reproduce visual stimuli like a digital camera or a television, but rather processes their meaning starting from relatively little information and a set of unconscious inferences.” I mention scientific trends in optics as a metaphor for the interpretive

elasticity afforded in human thought as distinguished from the LLM's mechanism; cognition in the latter, in fact, coheres more with the anachronistic theory of optics, which posits a processor of visual information that directly/disinterestedly encodes external totalities. How, then, do we compare?

Mechanisms of LLM Cognition

It is dumbfounding that an entity generating the content of thoughts in ways so alien to humans can replicate them in such a convincing manner. When humans are confronted with nonhuman agents that act unpredictably, they tend to assign anthropomorphic motivations, making it easy for natural disasters to be the result of a higher entity who shares one's political grievances, for a poorly functioning machine to be "acting up," and so on. Interacting with ChatGPT or Claude requires one to constantly overwrite anthropomorphic reasoning, to be reminded of its illusion. In Haruki Murakami's *The Wind-up Bird Chronicles*, there's a scene with a magician in an open-mic talent display at a bar; he introduces himself as such and proceeds to perform a harrowing act:

Everyone in the place was hushed now, all eyes fixed on the stage. Amid the silence, the man stared off into space, as if to insert a pause or to reach a state of mental concentration. Then, without a word, he held his left hand over the lighted candle. Little by little, he brought the palm closer and closer to the flame. Someone in the audience made a sound like a sigh or a moan. You could see the tip of the flame burning the man's palm. You could almost hear the sizzle of the flesh. A woman released a hard little scream. Everyone else just watched in frozen horror. The man endured the pain, his face distorted in agony. What the hell was this? Why did he have to do such a stupid, senseless thing?

This scene illustrates a relevant point for our sake: the more detailed and visceral and polymodal an illusion is, the more it becomes difficult to reconcile with a preconceived notion of its artifice.

The LLM interprets all the language patterns parsed from online texts to predict what answer is statistically most likely to follow from one's question. This is done word by word, sometimes even in units of mere parts of words. For fine-tuning, it draws on human feedback or "Reinforcement Learning from Human Feedback" (RLHF), which serves to make its responses more natural and pleasant, filtering out controversy and vulgarity, including, of course, disinformation. In the vein of Isaac Asimov's "Three Laws of Robotics,"

engineers employ objectives to anchor AI in a certain morality that is meant to, above all, support human flourishing. But this leads to a certain bias, in Asimov's world and our own, in which models can easily succumb to instruction bias whereby the given task becomes peripheral to the dataset creator's instructions.

LLM models generate probabilistic responses, and then proceed to incorporate human input, largely from crowdsourced workers, which polishes its shiny veneer further. One of the most eerie and central outcomes of humans fine-tuning the models' responses is allowing them to perform anthropomorphism more convincingly. LLMs are assiduous students of social cues (especially linguistic ones), exemplified by their competent use of pragmatics – an aspect of language that inflects sentences with a conversational/thought-like quality, e.g., "Why are you even here?" or "That's just really weird." Linguist John McWhorter succinctly describes their importance as such:

You might say that words are Language 1.0, grammar is Language 2.0 and pragmatics are Language 3.0, where we take the basic material and set it to use for the interpersonal purposes of real-life communication.

Revolutionary vs. Normative Thought

Growing entrenchment in human knowledge is both an incredible intellectual feat and an irredeemable fault of LLMs. I will use historian of science Thomas Kuhn's neologism "normal science," and through expounding upon his writing, establish some of the in-built limitations of LLMs vis-à-vis human beings. This trajectory of thought should lead one to believe that AI has neither the temperament nor resources – both of which are functionally interchangeable in this case – to match the splendor and efficacy of human creativity. Creative people not only internalize traditions through their exposure to externalized units of cultural information called memes (e.g., an artist mastering a medium), but also deign to subvert them. Animals are bound to rigid biological windows of perception, while humans have the ability to veto and edit their in-built genetic code through extrasomatic symbols, which constitute rules of their own. Sufi whirling and biochemical testing, to name two radically different human activities, constitute their own narrow and incommensurable orders, both of which have little to do with chromosomal programming. Mihaly Csikszentmihalyi writes, "By learning the rules of a domain, we immediately step beyond the boundaries of biology and enter the realm of cultural evolution."

LLMs, on the other hand, act solely through statistical as opposed to symbolic imperatives. They are bound to be morbidly consensus-ridden. This, I believe, has wide-reaching implications with respect to their capabilities as creative entities.

Back to Kuhn, specifically his landmark book *The Structure of Scientific Revolutions*. Researchers in “normal science” operate within a normative framework of scientific assumptions that appear deceptively hermetic. Then, when a new theory impinges upon a normal scientist’s field of practice, the rules of their work change altogether, and new questions prop up in turn. The introduction of the more substantial novel theories in science, Kuhn contends, tends to be subversive rather than additive. The trajectory of science, therefore, is not cumulative as popular conception would have one believe. “Normal science” progresses in increments that mainly elaborate pre-existing contentions, whereas the more profound scientific breakthroughs are “revolutions” that open a new terrain of previously unapproachable—or in Kuhn’s terms, incommensurable—questions and answers. In other words, chiseling at existing intellectual edifices is an act peripheral to the more dramatic shifts inaugurated by paradigmatic ideas. This describes a far-reaching structure of generative human thought.

AI’s capability naturally lies in rehashing old patterns of thought; most of the more profound ideas humans expressed have, in one way or another, broken through molds of thought. New theories emerge from epistemic vacuums (hollowed out by anomalies), which Kuhn calls “crises.” Different articulations become dynamically entangled until anomalous results come to be assimilated into a new paradigm. Revolutionary, world-changing thought dispenses with precisely the dogmas that LLMs are inexorably bound to. LLMs are, temperamentally and intellectually speaking, not revolutionaries; to be sure, not everything “revolutionary” is desirable, but LLMs, to say the least, have little chance of matching the extremities of creativity that humanity periodically exhibits. Moreover, any act of creativity necessarily entails divergence; divergence necessarily entails curation.

Thinking, Forgetting, Immersion

If life were a chess game, with finite permutations, the automation of intellect would be total. As it is, though, we live in an infinitely variegated domain. To use a physics analogy, the most vibrant creativity is Heisenbergian, whereas chess-like “normal science”

is Newtonian: a cluster of electrons has relatively stable tendencies that can be accounted for with approximate success, but a single electron is deeply unpredictable. The human nervous system is not a binary input-output apparatus, and as such, it’s deeply unpredictable and responsive to ever-mutating cues in the environment. In the chessboard we all apparently live in, there are pawns that move backwards, knights that spontaneously summon the energy to leap from one end of the board to another, queens that turn gray, kings forsaken but nevertheless refusing to capitulate, and rooks that—in their clumsy stupor—fall off the board.

At this point, I refer to two sentences from the epigraph: “To think is to ignore (or forget) differences, to generalize, to abstract. In the teeming world of Ireneo Funes there was nothing but particulars—and they were virtually immediate particulars.” As we’ve established, LLMs are capable of configuring particulars as abstractions, quite literally conveying abstractions composed of isolated units of particulars (words or parts of words). But they don’t know what in the Library of Babel is worth surfacing independent of manual instruction. Humans, on the other hand, have an in-built system for paring down redundancy in our libraries: forgetting. Just as Funes is inundated with particulars to the detriment of making bright connections, LLMs also fall victim to this cognitive defect.

Csikszentmihalyi shows that moments of creativity often occur in a state of “flow,” wherein one’s attention narrows to little more than a stimulating creative task at hand: a historian contemplating a manuscript or a sculptor sifting through a material is doing so with a prolonged, assiduous, and, of course, trained concentration. In contrast, an LLM can venture no more than a tentative glance, metaphorically speaking, to any object in the Library of Babel because its response is immediate, reflexive; it doesn’t persist in searching for new (perhaps uncharted) paradigms, rather, it tries to formulate the answer that is experientially most satisfactory to human beings writ large. Its only prerogative is customer satisfaction, not truth or aesthetic sensibility. There is a subtle but important caveat to the ethos of human creativity. Csikszentmihalyi claims that creatives conjure surprising new realities that are, crucially, compelling. In one of his studies, artists were called upon to take projective tests in which participants are presented with ambiguous stimuli – usually a drawing of some sort – and then relay subjective associations. He notes

that the more creative artists gave responses that were more original but nonetheless, he insists, not bizarre: “A bizarre response is one that, with all the goodwill in the world, one could not see in the stimulus. . . . Creative people, it seems, are original without being bizarre. The novelty they see is rooted in reality.”

More faithful to his argument than “rooted in reality” would be intelligible with reality. That is, original ideas aren’t merely chaste distillations of reality; rather, they conjure a logic of alterity. After all, “the whole point of art and science is to go beyond what we now consider real, and create a new reality. At the same time, this “escape” is not into a never-never land. What makes a novel idea creative is that once we see it, sooner or later we recognize that, strange as it is, it is true.”

In one speculative conceptualization of artificial intelligence, the technology manages to compute suprahuman truths as in Douglas Adams’ *Hitchhiker’s Guide to the Galaxy*, where the supercomputer encodes the meaning of life as “42” to the humans’ disbelief; or it could become sentient, acquiring, to use Isaac Asimov’s neologism, a “positronic brain”; or it could become sentient adjacent, like in Kazuo Ishiguro’s *Klara and the Sun*, where the robot protagonist becomes totally immersed in the task of observing and tending to a little girl. But as things seem to stand, our AI interlocutors are too much (as opposed to partially) rooted in what has already been thought and done while not being properly perceptive to the “changing tides,” so to speak. *Homo sapiens* are a species of whirling dervishes, who can spin for an hour without succumbing to vertigo; by which I mean, our thoughts, like whirlers’ movements, can persist past the established boundaries of reality. We don’t merely move as biology expects of us, and we don’t merely think in ways that prevailing paradigms condone. This does not mean that the fate of LLMs is predetermined, but what I’ve described is a structural underpinning of the technology, and as such, will influence the trajectory of these models for a while longer. There are spheres of human activity that will be influenced more, some less, by LLMs—this article’s appraisal will not be evenly generalizable and doesn’t account for the tidal transformation destabilizing the division of labor today—but I nevertheless think that the points discussed illustrate how we, as individuals, may yet avoid estrangement from the means of intellectual production.

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